

Chapter 2

Contours

Class 10 - Total Geography Morning Star

Give the terms used in contours for each of the following

Question 1

A number preceded by a dot.

Answer

Spot height

Question 2

The numerical difference between two contour lines.

Answer

Vertical interval

Question 3

A number preceded by a triangle.

Answer

Triangulation point

Question 4

The difference between maximum and minimum height in areas of steep slope.

Answer

Relative height

Question 5

The accurate height of some point marked above the ground.

Answer

Spot height

Describe the following terms

Question 1

Contour Interval

Answer

Contour interval means the difference of height between two consecutive contours. For example, if one thick contour shows the height 300 metres and the next thin contour depicts the height of 320 metres. Then, the contour interval is 20 metres.

Question 2

Vertical Interval

Answer

The vertical distance between any two contour lines is called vertical interval.

Question 3

Horizontal Equivalent

Answer

The actual distance between two points on two contour lines is called horizontal equivalent.

Question 4

Gradient

Answer

The ratio between the vertical height and the horizontal distance of a slope of land measured along the ground is called gradient.

Explain how would you distinguish between the following on the basis of contours

Question 1

Gentle Slope and Steep Slope

Answer

Gentle Slope	Steep Slope
In gentle slopes, the change of elevation is gradual. Normally about 1:25 gradient of elevation is gradual.	Steep slope suddenly slopes
The contours are placed wide apart in gentle slope.	For steep slopes, the contours are close to one another.

Question 2

Scarp and Dip

Answer

Scarp	Dip
A scarp has a narrow top with a steep slope.	A dip has a narrow top with a gentle slope.
The contour lines are closer to each other.	The contour lines are spread far apart.
It has a gradient of 1:3.	It has a gradient of about 1:12.

Question 3

Ridge and Water Divide

Answer

Ridge	Water Divide
It is a long and narrow highland sloping steeply downwards on its sides.	A line which separates two adjacent drainage basins is known as a water divide.

Ridge	Water Divide
Contours showing a ridge are elongated and are closely spaced.	The side on which the ridge is steeper, the contours are drawn close together, whereas the side on which the ridge is gentle, the contours are drawn far apart.

Question 4

Peak and Depression

Answer

Peak	Depression
The highest point of a mountain is known as a mountain peak.	A depression is formed when there is a sudden decrease in elevation at the top of a conical hill.
Contours showing a conical mountain are almost circular with peak of the mountain as their centre. The contours are drawn almost at regular distances, the value of which increases towards the centre, which is the peak and represented with a dot.	Contours showing a depression are almost circular with a depression as their centre. The contours are drawn almost at regular distances, and their value keeps on increasing towards the centre, where the lines suddenly show decrease in elevation, which is the depression.

Short Answer Questions

Question 1

What is meant by 'relief'?

Answer

Relief refers to actual configuration of land which includes its altitude and slope. The relief features found on the earth's surface include mountains, valleys, hills, plains and plateaus.

Question 2

State the three methods used to show the earth's relief features on a map.

Answer

The three methods used to show the earth's relief features are-

1. hachures

2. hill shading
3. contours

Question 3

What is a 'contour'?

Answer

Contour lines are imaginary lines which are drawn on a map to join places having the same height above the mean sea level.

Question 4

What do the contour lines represent?

Answer

Contour lines are imaginary lines joining places which have the same height above mean sea level.

Question 5

What advantages do contours have over hill-shading and hachuring?

Answer

Hill shading and hachuring only give an impression of the slope but do not indicate the heights of the land above sea level, whereas the contours show both impression of the slope as well as the heights of the land above the sea level.

Question 6

How do contour lines show different slopes of land?

Answer

The contour lines are shown by brown colour as thick brown lines and thin brown lines. Contour lines show:

1. Elevation of land: The height of a contour is indicated by figures inserted in a break in the line.

2. Steepness of its slope: When the contours are very close, they represent steep slopes. When they are farther apart, they represent gradually increasing slopes. Absence of contour lines indicate that the land is flat i.e., a low land.

3. Shape of land at various heights: The contour lines indicate the shape of land. For example, almost circular contours whose value decreases inside represent a lake.

Question 7

What do you understand by V.I. and H.E.?

Answer

V.I. stands for vertical interval. It is the vertical distance between any two contour lines.

H.E. stands for horizontal equivalent. It is the actual distance between two points on two contour lines.

Question 8

Give two points of difference between Bench Mark and Spot Height.

Answer

Bench Mark	Spot Height
These are marks etched on stone or a building to indicate the accurate height determined by surveys.	It refers to the height of a point on the mean sea level.
It is written as "BM" followed by a number.	It is represented by a dot followed by the map.

Question 9

Why do the contour of different heights not cross each other?

Answer

Two contours never intersect each other because each line represents a different height, and we can't have two different heights at the same point.

Question 10

How do the contour lines show the direction of the flow of rivers and streams?

Answer

Rivers and streams are shown on the map with blue colour. Generally water flows from higher altitude to lower altitude. Since the contours indicate the elevation of land, we can use them to find the direction of a river or stream.

The rivers and streams usually flow from contours representing greater elevations to contours representing lower elevations. The course of a well-developed river can be divided into three stages - upper, middle and lower course; each of which can be recognised with the help of contour lines.

Symbols and Conventional Signs

Question 1

Given below are some symbols and conventional signs, name each one:

Symbol/Conventional Sign	Name	Symbol/Conventional Sign	Name
1.	Kiln	7.	
2.		8.	
3.		9.	
4.		10.	
5.		11.	
6.		12.	

Answer

1. Kiln
2. Tank with Embankment
3. Permanent huts
4. Temple
5. Lighthouse
6. Triangulated station
7. Benchmark
8. Post office
9. Overhead tank
10. Police station
11. Hospital
12. Powerline with pylons surveyed

Question 2

Complete the following by drawing the conventional sign/symbol in the blank boxes:

Name	Symbol/Conventional Signs	Name	Symbol/Conventional Signs
1. Cart-track		Pack-track with pass	
2. Dam (masonry)		Dams (earthwork)	
3. Dry river with water channel		River with island & rocks	
4. Lined well		Unlined well	
5. Perennial well		Dry well	
6. Permanent Hut		Temporary Hut	
7. Temple		Chhatri	
8. International boundary		State boundary	
9. Inspection bungalow		Circuit house	
10. Hospital		Dispensary	

Answer

	Name	Symbol/Conventional Signs	Name	Symbol/Conventional Signs
1.	Cart-track		Pack-track with pass	
2.	Dam (masonry)		Dams (earthwork)	
3.	Dry river with channel		River with island and rocks	
4.	Lined well		Unlined well	
5.	Perennial well		Dry well	
6.	Permanent hut		Temporary Hut	
7.	Temple		Chhatri	
8.	International boundary		State boundary	
9.	Inspection bungalow		Circuit house	
10.	Hospital		Dispensary	

Project / Activity

Question 1

Draw conventional symbols for the following using appropriate colours where necessary.

(i) Cart-track

(ii) Broken ground

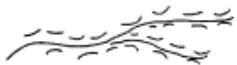
(iii) Nala

(iv) Perennial tank

(v) Stream

(vi) Dry-stream

Answer

	Name	Symbol		Name	Symbol
1.	Cart-track		2.	Broken ground	
3.	Nala		4.	Perennial tank	
5.	Stream		6.	Dry-stream	